



Multi-task Multi-instance Learning for Jointly Diagnosis and Prognosis of Early-Stage Breast Invasive Carcinoma from Whole-Slide Pathological Images

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Abstract. With the tremendous progress brought by artificial intelligence, many whole-slide pathological images (WSIs) based machine learning models are designed to estimate the clinical outcome of human cancers. However, most of the existing studies treat the prognosis and diagnosis tasks separately, which overlooks the fact that the diagnosis information indicating the severity of the disease that is highly related to the patients' survival. In addition, it is still challenge to design machine learning models to analyze WSIs since a WSI is of large size but only annotate with coarse labels, it increasingly becomes a research hotspot for the development of automated WSI analysis tools in a scenario without fully annotated data. Based on the above considerations, we propose a multi-task multi-instance (WSI-MTMI) learning method that can simultaneously conduct the prognosis and diagnosis tasks from WSIs. Specifically, inspired by the fact that taking the associations between diagnosis and prognosis tasks can improve the generalization ability for each individual task, we firstly adopt multi-instance learning algorithms to aggregate the patches derived from WSIs by considering both the common and specific task information. Then, we design a novel cross-attention network that can effectively identify useful information shared across different tasks. To evaluate the effectiveness of the proposed method, we test it on the early-stage breast invasive carcinoma (BRCA) derived from the Cancer Genome Atlas project (TCGA), and the experimental results indicate that our method can achieve better performance on both diagnosis and prognosis tasks than the related methods.

Keywords: Multi-task Multi-instance Learning · Pathological images · Cancer diagnosis · Cancer prognosis

1 Introduction

Cancer is the second leading cause of death worldwide. It is reported that the number of affected people has already reached to a high record of 19.3 million

in 2020 [13]. Thus, the effective and efficient analysis and intervention of human cancer, especially at its early stage, is crucial in treatment selection and planning for each cancer patient.

So far, amounts of bio-markers have been developed and applied to analyze human cancer, including the histopathological images, genetic mutations, epigenetic signatures, and protein markers. Of all these types of biomarkers, the pathological examination remains to be the gold standard for the clinical outcome prediction of human cancers since the whole-slide histopathological images (WSIs) can reveal the degree of tissue cancerization by considering the morphology and distribution of cells within them. Recently, the emerging of digital pathology makes computerized quantitative analysis of histopathology imagery possible, and many computational models have been employed for the prediction of human cancers.

Generally, the existing WSI-based predictions models can be generally divided into two folds i.e., diagnosis and prognosis models. The diagnosis models aim at building classification models to predict the categorical variables (e.g. TNM staging and Nottingham Histologic Score) from WSIs. For example, Zhao et al. [18] designed a self-supervised learning algorithm to extract meaningful features from histopathological images, followed by applying the graph convolutional network (GCN) to drive diagnostic practice. Zhang et al. [17] built a double-tier multi-instance learning framework that can localize the tissue regions to help cancer diagnosis. Moreover, Gao et al. [2] proposed an instance-based Vision Transformer to learn robust representations of histopathological images for the subtype classification task.

On the other hand, besides the diagnosis tasks, many prognosis models were also adopted to perform survival analysis of human cancers on WSIs. For instance, Katzman et al. [4] introduced a Cox proportional hazards deep neural network for modeling interactions between a patient's covariates and treatment effectiveness. Yao et al. [16] proposed Deep Attention Multiple Instance Survival Learning by introducing both Siamese MI-FCN and attention-based MIL pooling to efficiently learn imaging features from the WSI and then aggregate WSI-level information to patient-level. Lv et al. [7] developed a multi-scale histopathological features fusion transformer (MS-Trans) to integrate the relationship of different image patches for survival analysis.

Although much progress has been achieved, most of the existing methods conduct the diagnosis and prognosis tasks separately, without considering their inherent correlation. As a matter of fact, the diagnosis information indicates the extent of the disease severity, which is highly correlated with the patients' clinical outcomes. For instance, the patients in Stage II suffer from more aggressive cancer than those in Stage I, and thus generally with shorter survival time [11]. It is expected that better diagnosis and prognosis performance can be achieved if we take the association between them into consideration. In addition, due to the high resolution of WSI, it is impossible to directly feed the WSIs into the neural network for model training. In recently years, many multi-instance learning (MIL) algorithms focusing on designing proper aggregation strategies to fuse the patch-level prediction results to WSI-level have been proposed to

deal with such challenges. However, those MIL methods simply used the pooling strategies or consider the associations among different patches for patch fusion, which overlooks the similarity and variability of different tasks if we directly extend the existing MIL algorithm to multi-task learning scenario.

Based on the above consideration, we propose a Multi-task Multi-instance learning framework for jointly diagnosis and prognosis of early-stage breast invasive carcinoma from whole-slide pathological images. Specifically, basing on the self-attention mechanism, the task similarity in multi-instance learning is captured by sharing a common set of value vectors derived from the sampled patches across all tasks. In addition, our method also extracted the specific key and query vectors from the instance patches, which can capture the variability of the patches across different tasks. Then, we adopt the multi-head attention algorithm to generate the sample-level embedding from the patches. Finally, a novel cross-attention network that can effectively identify useful information shared across different tasks is applied to conduct the diagnosis and prognosis tasks, simultaneously.

To evaluate the effectiveness of the proposed method, we test it on the early-stage breast invasive carcinoma (BRCA) derived from the Cancer Genome Atlas project (TCGA), and the experimental results indicate that our method can achieve better performance on both diagnosis and prognosis tasks than the related methods.

2 Method

We now introduce the proposed WSI based multi-task multi-instance learning architecture (WSI-MTMI) shown in Fig. 1, which can be roughly divided into two steps, 1) extracting patch-level histopathological imaging features and aggregating them into sample-level representations via multi-instance learning module by considering the common and specific task information; and 2) fusing layer features via cross-attention mechanisms which identify useful information shared across different tasks for conducting the diagnosis and prognosis tasks on cancer patients, simultaneously.

Multiple Instance Learning (MIL). Generally, multi-instance learning (MIL) methods take the instances derived from a sample as inputs, $X_i = \{x_1^i, \dots, x_N^i\}$, and predicts the sample-level label Y_i . In MIL, instance-level label y_n^i is usually unknown for model training. As for a binary classification task, sample-level label Y_i is positive only if there are at least one positive instance and labelled as negative when none of them is positive, which could be formulated as:

$$Y_i = \begin{cases} 0, & \text{if } \sum y_i = 0, \quad i = 1, \dots, n \\ 1, & \text{otherwise} \end{cases} \quad (1)$$

Furthermore, MIL uses a scoring function $S(X)$ to predict the Y_i :

$$S(X) = g \left(\sum_{x \in X} f(x) \right) \quad (2)$$

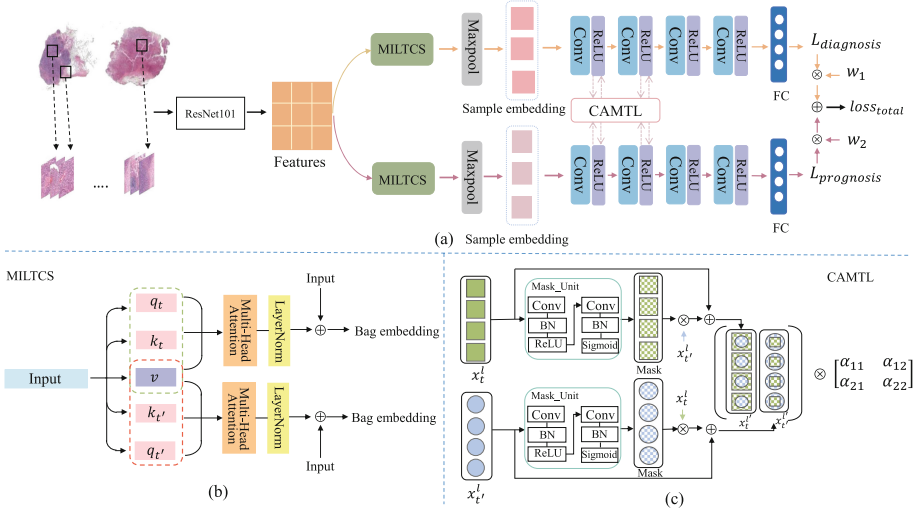


Fig. 1. The overall framework of the WSI-MTMI network (Fig. 1(a)). First, patch-level representation extracted from the pre-trained Resnet-101 network are aggregated by the proposed multi-instance learning module (Fig. 1(b)), which considers the common and specific task information (MILTCS) for multi-task learning; Second, a novel cross-attention network (CAMTL) that can effectively identify useful information shared across different tasks is applied to conduct the diagnosis and prognosis tasks, simultaneously (Fig. 1(c)).

where f is the non-linear transformation operated on the instances level and g is an aggregation operator at the sample level. In this work, we take the full account of the intrinsic relationship among instances for the multi-instance learning.

Multi-instance Learning by Considering the Common and Specific Task Information (MILTCS). In this work, multiple instance learning is casted into a multi-task learning setting, where effective information can be separated into the task-specific and task-shared features across various tasks. In view of this, we develop a self-attention based multi-instance learning module, which captures the common information by a shared value sets while preserves task-specific information by the respective query and key value sets.

As shown in the Fig. 1(b), the input sample of feature $E_m = \{e_{m,1}, \dots, e_{m,n}\} \in R^{d \times n}$ is transformed into task-specific query (key) values q_t (k_t) with the weight matrix W_{qt} (W_{vt}) respectively, and the common value sets v shared across tasks. It is worth noting that the utilization of common value sets encourages the model to exploit the complementary supervision information from different tasks. Meanwhile, this manner also reduces the computation complexity for the

sample-level representation learning. For the t_{th} task, self-attention mechanism is implemented as:

$$\begin{aligned} q_t^h &= W_{qt}^h \cdot E_m \\ k_t^h &= W_{kt}^h \cdot E_m \\ SA_t^h &= \text{softmax} \left(\frac{q_t^h \cdot k_t^{h^T}}{\sqrt{d/h}} \right) \end{aligned} \quad (3)$$

where h is the number of heads in self-attention module. The sample-level embedding B_m^t is calculated as:

$$\begin{aligned} v_{\text{shared}}^h &= W_{v-\text{shared}}^h \cdot E_m \\ \text{head} &= SA_t^h \cdot v_{\text{shared}}^h \\ MHA(q, k, v)_t &= \text{Concat}(\text{head } t, \dots, \text{head } t) W^o \\ B_m^t &= \text{Maxpool}(MHA(q, k, v) + E_m) \end{aligned} \quad (4)$$

where head_t^h is self-attention of head h , while W^o refers to the multi-head attention weight matrix. We aggregate the patch-level features into bag-level representation via max pooling operation.

Different from previous studies [12], we discard the multi-layer perception layers in the proposed MILTCS module and extend to the multi-task learning setting by specifying the common value sets. Meanwhile, task-specific query and key sets are expected to concentrate on distinct information of different tasks.

Cross-attention Multi-task Learning (CAMTL). As described above, our MILTCS module explicitly models the shared and task-specific information to obtain the sample-embedding. In order to more effectively fuse the features derived from different tasks, as shown in Fig. 1, we propose a novel cross-attention multi-task learning (CAMTL) module that can effectively identify useful information shared across different tasks. We implement such feature fusion operation at multiple layers. Figure 1(b) presents the details of this module consisting of the mask generation module and cross-stitch module [9]. The main motivation behind is that cross-attention mechanism enables to further exploit the collaborative effect of different tasks. The generation of the mask for the opposite task mask_t^l is inspired by [6], We formulate this unit as:

$$\text{mask}_t^l = g_t^l(h_t^l(x_{t'}^l)) \quad (5)$$

where x_t^l represents the output of task t at layer l . Like in [6], both h_t^l and g_t^l are composed of 1×1 kernels with batch normalization, followed by a non-linear activation. The mask_t^l is restricted in $[0-1]$. We could figure that the higher value means that $x_{t'}^l$ has the larger effect on x_t^l at certain dimension. $\text{mask}_t^l \rightarrow 1$ equals to $p_t^l = x_t^l + x_{t'}^l$, and $\text{mask}_t^l \rightarrow 0$ reduces to an identity map, i.e., no effect imposed on task t from t' . Clearly, this manner provides us a flexible way to extract useful information during feature fusion.

The features of task t are re-calculated via element-wise multiplication between the mask mask_t^l and the features $x_{t'}^l$, and then fused with the raw feature x_t^l by a residual connection. Finally, followed by the cross-stitch module [9], we can derive:

$$\begin{bmatrix} p_t^{l+1} \\ p_{t'}^{l+1} \end{bmatrix} = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \cdot \begin{bmatrix} \left[\text{mask}_t^l \otimes x_{t'}^l \right] + x_t^l \\ \left[\text{mask}_{t'}^l \otimes x_t^l \right] + x_{t'}^l \end{bmatrix} \quad (6)$$

$$\begin{bmatrix} x_t^{l+1} \\ x_{t'}^{l+1} \end{bmatrix} = \begin{bmatrix} g_t^{l+1} (p_t^{l+1}) \\ g_{t'}^{l+1} (p_{t'}^{l+1}) \end{bmatrix} \quad (t \neq t') \quad (7)$$

where p_t^{l+1} is the input of the task t at layer $l+1$, $\otimes$ refers to the element-wise multiplication, α_{ij} is the weight of linear combination, g_t^l is composed of 3×3 kernels, followed by downsampling module to reduce the resolution.

Loss Function. For diagnosis task, we adopt mean square error as loss function as:

$$l_{\text{diagnosis}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (8)$$

where the y_i is the ground truth label, $\hat{y}_i$ is the prediction of diagnosis task.

As to prognosis task, the Cox proportional hazard model is applied to predict the patients' clinical outcome [11], and its negative log partial likelihood function can be formulated as:

$$l_{\text{prognosis}} = \sum_{i=1}^N \delta_i \left(\theta^T x_i - \log \sum_{j \in R(t_i)} \exp(\theta^T x_j) \right) \quad (9)$$

where x_i represents the output of the last layer for the prognosis task and $R(t_i)$ is the risk set at time t_i , which represents the set of patients that are still under risk before time t . In addition, δ_i is an indicator variable. Sample i refers to censored patient if $\delta_i = 0$, otherwise $\delta_i = 1$ patient.

By combining Eq. (8) and Eq. (9), the final objective function for the proposed WSI-MTMI method can be formulated as:

$$l_{\text{total}} = l_{\text{diagnosis}} + \lambda \times l_{\text{prognosis}} \quad (10)$$

where λ is the weight of prognosis task loss.

3 Experiments

To verify the superior performance of our model on the diagnosis and prognosis of cancer patients, we test it on early-stage breast invasive carcinoma (BRCA) dataset public available on The Cancer Genome Atlas (TCGA) database. We show the demographic information of the utilized dataset in Table 1.

Experimental Setting. For each WSI, we randomly sampled 100 non-overlapping adjacent cancer patches with the size of 512×512 from each WSI. Then, we applied the five-fold cross validation strategy to evaluate the performance of our method. Specifically, we iterative selected the patients in one fold as the testing data at each time, while the remaining patients are used for model training. From the training set, we randomly partition 20% samples as validation set to tune model parameters. To fairly compare the prediction results among different methods, we applied the same way to split training and testing dataset. For each patch, we followed the method in [19] that applied the pre-trained ResNet-101 network to extract a 2048-dimensional vectors, followed by applying PCA to reduce its dimension to 1000. The λ is tuned from [1.5, 2.0]. The CAMTL module is consisted of four convolutional layers, and the cross-attention operations are applied on the first two layers. We set the learning rate as 0.07 during the training process of the WSI-MTMI model. We evaluate the performance of the diagnosis task by the measurements of Accuracy, F1-Score, Recall and Precision. For the prognosis task, we take the Concordance Index (according to Eq. (9), $CI = \frac{1}{n} \sum_{i \in \{1..N | \delta_i=1\}} \sum_{t_j > t_i} I(x_i \theta > x_j \theta)$) and AUC as evaluation metrics. Both the value of CI and AUC range from 0 to 1. The larger CI and AUC value mean the better prediction performance of the model and vice versa.

Evaluate Our WSI-MTMI Method on Diagnosis Classification. In this section, we compare the diagnosis power of the proposed WSI-MTMI with the following five algorithms on BRCA dataset by the measurements of Accuracy, F1-Score, Recall and Precision: (1) DSMIL [5]: A single-task method which designs a dubbed dual-stream multiple instance learning network for WSI classification. (2) Graph Attention MIL (GATMIL) [10]: A single-task method based on Graph Attention MIL for TNM staging prediction. (3) WSI-MID: A variant of the proposed method, which (4) MTAN [6]: A MTL method which using a shared feature network, each task select task-specific feature through attention networks. (5) Cross-stitch [9]: Applying the cross-stitch operation to get shared representations for multi-task learning. (6) NDDR-CNN [1]: A MTL method which apply discriminative dimensionality reduction technique for layer-wise feature fusion across different tasks. The classification results are shown in Table 2.

From Table 2, we can derive the following observations: (1) The multi-task learning algorithms (i.e., MTAN, Cross-stitch and NDDR-CNN) can consistently

Table 1. Demographics and Clinical Characteristics of the Utilized Dataset

Characteristics	Summary	Characteristics	Summary
Patients		Stage	
Censored	571	StageI	125
Non-censored	47	StageII	493
Age(Y)	58.1(± 12.9)	Follow-up(M)	27.1(± 33.9)

Table 2. The performance of different methods on diagnosis task

Method	Accuracy	F1-Score	Recall	Precision
GATMIL [10]	0.705(± 0.039)	0.782(± 0.033)	0.791(± 0.043)	0.770(± 0.037)
WSI-MID	0.709(± 0.026)	0.800(± 0.027)	0.755(± 0.060)	0.826(± 0.020)
DSMIL [5]	0.714(± 0.025)	0.817(± 0.026)	0.808(± 0.067)	0.829(± 0.031)
MTAN [6]	0.727(± 0.044)	0.829(± 0.037)	0.841(± 0.083)	0.821(± 0.025)
Cross-stitch [9]	0.733(± 0.037)	0.832(± 0.040)	0.815(± 0.092)	0.807(± 0.090)
NDDR-CNN [1]	0.736(± 0.045)	0.830(± 0.043)	0.774(± 0.090)	0.832(± 0.052)
WSI-MTMI	0.772(± 0.038)	0.857(± 0.035)	0.870(± 0.091)	0.848(± 0.031)

Table 3. The performance of different methods on prognosis prediction task

Method	C-index	AUC
DeepSurv [4]	0.650(± 0.022)	0.700(± 0.083)
Cross-stitch [9]	0.665(± 0.033)	0.762(± 0.088)
DeepAttnMIL [16]	0.698(± 0.018)	0.724(± 0.071)
NDDR-CNN [1]	0.698(± 0.016)	0.751(± 0.022)
MTAN [6]	0.701(± 0.021)	0.764(± 0.024)
WSI-MTMI	0.721(± 0.025)	0.785(± 0.041)

achieve more accurate discrimination than the single-task algorithms (i.e., GAT-MIL, DSMIL). These results clearly verify the advantage of multi-task learning that can combine information from other tasks in order to obtain better features for cancer diagnosis. (2) The proposed WSI-MTMI can achieve the accuracy of 0.772 and the F1-score of 0.857, which is superior to existing MTL learning methods (i.e., MTAN, Cross-stitch and NDDR-CNN). These results validate the superiorities of capturing the correlation between diagnosis and prognosis tasks can help promote the diagnosis performance.

Evaluate Our WSI-MTMI Method on Prognosis Prediction Task. In this section, we evaluate our proposed method on prognosis prediction task. In addition to the three multi-task methods (i.e., MTAN, Cross-stitch and NDDR-CNN) discussed in the previous section, we also compare the proposed method with two single-task prognostic methods. (1) DeepSurv [4]: a Cox proportional hazards deep learning method. (2) DeepAttnMIL [16]: an attention based MIL method to aggregate WSI-level feature to patient-level for survival learning. We also test the performance of our method on prognosis prediction task. The experimental results are shown in Table 3.

As can be seen from Table 3, WSI-MTMI consistently outperforms the comparing methods on prognosis prediction task. Specifically, our proposed method WSI-MTMI achieves the best average Concordance Index (C-index) of 0.721 and the AUC of our method is also superior to its competitors. These results clearly

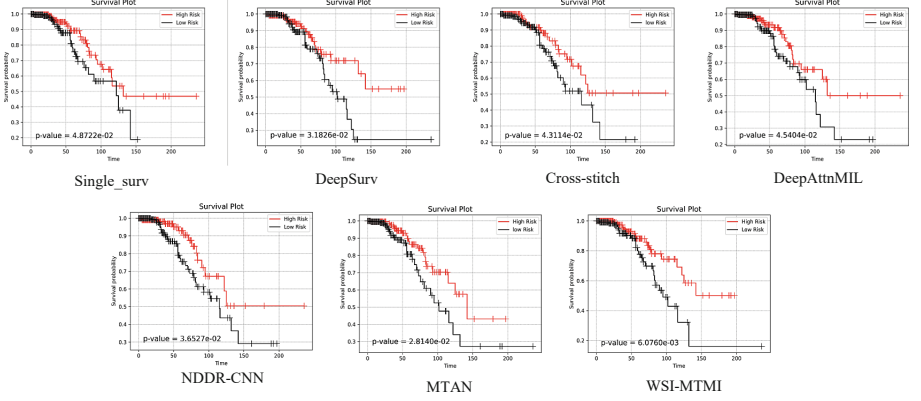


Fig. 2. The survival curves by applying different methods on BRCA datasets

demonstrate that the incorporation of the prior diagnosis information (i.e., TNM stage) can help improve the prognosis performance. In addition, both the experimental results shown in Table 2 and Table 3 indicate that the diagnosis and prognosis tasks are highly correlated with each other, and applying the multi-task learning framework to capture the correlation between different tasks can promote the prediction performance for each individual task.

Evaluate Our WSI-MTMI Method on Patient Stratification Task. In the field of prognosis prediction, another important task is to stratify cancer patients into subgroups with different predicted outcomes, by which we could intervene with personalized treatment plan in the stage of cancer development. In this section, we also compare the prognostic power of different approaches by stratifying cancer patients into 2 subgroups (i.e., the high and low survival risk groups) with different predicted outcomes, with the experimental results shown in Fig. 2. Specifically, the stage method divides all the patients into 2 subgroups according to the TNM stage (i.e., stage I and stage II). For the proposed WSI-MTMI method and other competitors, Gaussian clustering algorithm is adopted to aggregate the patients into 2 subgroups based on the extracted features. Then, we test if these 2 subgroups has significantly different survival outcome using log-rank test [14]. As shown in Fig. 2, the proposed WSI-MTMI method can achieve superior stratification performance (i.e., with p-values of 0.006) than the comparing methods. These results further show the promise of applying task relationship learning framework to identify survival-associated feature shared across different tasks for patient stratification, which opens up the opportunity for building personalized treatment plan in the stage of cancer development.

The Effects of Applying Cross Attention Operations on Different Layers of CAMTL Module. Moreover, we also investigate the effect of applying

cross-attention operations on different layers of the CAMTL module, and the experiment results are shown in Table 4, where the number in the first column of Table 4 indicate which layers in the designed CAMTL module adopt the cross-attention operations.

Table 4. The Effects of Applying Cross Attention Operations on Different Layers of the Network

Strategy	Accuracy	F1-Score	Recall	Precision	C-index	AUC
No Cross-Attention	0.729 (± 0.078)	0.820 (± 0.072)	0.811 (± 0.015)	0.850 (± 0.044)	0.664 (± 0.048)	0.653 (± 0.069)
Cross-Attention(3,4)	0.677 (± 0.062)	0.780 (± 0.041)	0.732 (± 0.073)	0.835 (± 0.024)	0.673 (± 0.026)	0.717 (± 0.056)
Cross-Attention(1,2,3,4)	0.736 (± 0.060)	0.829 (± 0.046)	0.817 (± 0.084)	0.844 (± 0.022)	0.686 (± 0.042)	0.718 (± 0.079)
Cross-Attention(1,4)	0.750 (± 0.060)	0.838 (± 0.051)	0.832 (± 0.081)	0.850 (± 0.015)	0.689 (± 0.046)	0.742 (± 0.073)
Cross-Attention(1,2)	0.772(± 0.038)	0.857(± 0.035)	0.870(± 0.091)	0.848(± 0.031)	0.721(± 0.025)	0.771(± 0.022)

As can be seen from Table 4, the diagnosis and prognosis performance will be dramatically decreased if no cross-attention operations are applied, which again validate the fact that take the correlation between diagnosis and prognosis tasks can improve the performance of each individual task. In addition, it is noteworthy that applying the cross-attention operations only on the higher layers (i.e., layer 3 and 4) of CAMTL module will lead to inferior diagnosis and prognosis performance, this is because higher layers contain more task-related information, sharing such information will deteriorate the prediction results for each task. On the other hand, the prediction performance will be largely improved if applying the cross-attention operation on the low layer (i.e., layer 1 and 2) of CAMTL module, one possible reason is that the low-layer cross-attention operations can share common information that is beneficial to both tasks.

Significant Biomarkers Detected by the Proposed WSI-MTMI Method. In this section, we visualize the sample patches that are existed in both high and low survival risk groups in Fig. 3. The patches in high survival risk group (the first line) have large percentage of large nuclei than those in the low survival risk. These results are consistent with the existing discovery that large values of nuclei size usually comes with worse prognosis [15]. In addition, it is worth noting that the nuclei arranged more closer in high survival risk patients, which is in accordance with the fact that the higher tumor cell density will result in worse prognosis and vice versa [15].

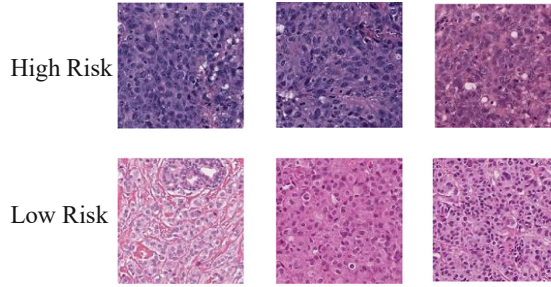


Fig. 3. Sample patches in both high and low survival risk groups.

4 Conclusion

In this paper, we propose a multi-task multi-instance learning for jointly diagnosis and prognosis of early-stage breast invasive carcinoma from Whole-slide pathological images. The main contribution of this study lie in the following two aspects. Firstly, we cast multiple instance learning into a multi-task learning setting and develop a self-attention based multi-instance learning module, which captures the common information by a shared value sets while preserves task-specific information by the respective query and key value sets. Secondly, we propose a cross-attention networks to effectively identify useful information shared across different tasks. Thus, it can avoid negative transfer due to the addition of other information. The experimental results indicate that our method performs better on both diagnosis and prognosis tasks than the existing studies.

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